

Smart University Management System

Enterprise Database Project Report

Architecture: Polyglot Backend + MySQL 8.0 | **Scale:** 5 Campuses, 50,000+ Students

1. Introduction: Why SUMS?

The Smart University Management System (SUMS) was built to solve the data fragmentation problem that large universities face. In a traditional setup, different campuses operate on different servers. This means administrators spend weeks compiling spreadsheets just to figure out basic metrics like student attendance, fee collection, or placement success.

SUMS is a centralized "Global Hub". It connects 5 distinct campuses into one single, powerful MySQL database. This allows over 50,000 student records to be managed seamlessly in real-time.

How It Helps the People

For the Students

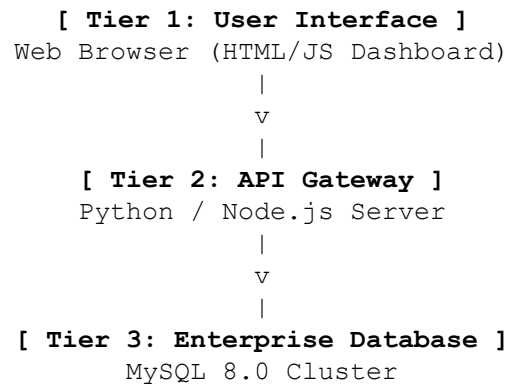
Students no longer need to wait in long lines to request transcripts, NOCs, or check their grades. They can log into the portal and request documents digitally. The system automatically sends these requests to the right department and prioritizes them based on urgency.

For the Administrators

University executives get a live dashboard. If they want to know the exact value of IT assets at the Chennai campus or the highest CTC offered by Google this year, they can see it instantly without asking for reports. Everything is automated.

2. How It Works: The Architecture

To make sure the system is secure and fast, it is divided into three separate layers. The user interface (what you see) never talks directly to the database. Instead, it uses an "API Gateway" as a secure middleman.



A Practical Example: Requesting a Document

When a student clicks a button to request a document, the system follows this simple path:

1. **Input:** The student selects "Transcript" and "Urgent" on the website.
2. **Command:** The website sends a message to the Python API.
3. **Storage:** The Python API securely writes this request into the MySQL database using an INSERT command.
4. **Output:** Later, when the Registrar opens their dashboard, the database sends back the list of pending requests, which the website draws as an organized table.

3. Database Schema & Smart Automation

The database uses over 30 interconnected tables to make sure no data is duplicated. This is called "normalization".

Primary Table	Purpose	Connections (Foreign Keys)
Campuses	Stores the 5 regional hubs.	Links to Departments and Assets.
Students	Stores 50,000+ academic profiles.	Links to Enrollments and Payments.
Enrollments	Maps students to the courses they take.	Links to Students and Course Sections.
PlacementResults	Tracks job offers and CTC packages.	Links to Students and Companies.

Smart Automation

Because there are so many students, we cannot rely on humans to calculate grades manually. The database does the math automatically using built-in logic:

- **Automated Grading:** When a professor enters a grade, a MySQL *Trigger* activates. It immediately recalculates the student's total credits and overall CGPA without anyone having to click a button.
- **Automated Graduation:** When it is time to graduate, an administrator runs a *Stored Procedure*. The database instantly moves the student's record to the Alumni table and archives their active status.

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